

Bhanu Pratap

Certified Databricks Data Engineer

Phone: 215-518-4468 | Email: bhanu.tcs@gmail.com | NJ, USA | [LinkedIn Profile](#) | [GitHub](#)

PROFILE

Certified Databricks Data Engineer with over 19 years of IT experience, including 5+ years specializing in the Databricks Lakehouse Platform. Expert in architecting large-scale ETL/ELT frameworks and Medallion data architectures using Python, SQL, and Spark. Proven track record of optimizing data pipelines to reduce processing latency and infrastructure costs in complex cloud environments. Delivered high-volume data platforms that power analytics and decision-making for Fortune 10 healthcare and retail clients. Skilled in optimizing performance and cost, enabling efficient data operations across batch and streaming workloads. Experienced in GenAI and ML proof-of-concepts, and adept at translating complex business requirements into scalable, reliable, and governed data architectures.

TECHNICAL SUMMARY

Programming Languages: Python, SQL

Big Data Technology: PySpark, Spark SQL

Streaming Technology: Spark structured streaming

Cloud Computing: Azure Data Factory (ADF), Databricks, ADLS, Amazon S3, AWS Lambda, SQS

Data Engineering Tools: Azure DevOps, Data Modelling, ETL/ELT data pipeline

Orchestration: Control-M, Databricks Workflows

Data Governance & Security: Unity Catalog

Familiarity: GCP

EXPERIENCE

Tata Consultancy Services

NJ, USA

Data Engineer and Architect

Project: Enterprise Data Lakehouse on Databricks

Client: CVS Health | **Duration:** Jan 2023 – Present (3 years)

Role : Data Engineer and Architect

Key Contributions:

- Architected and delivered a robust Data Lakehouse, designing end-to-end pipelines in **Databricks** using **PySpark** and **SQL** to support high-volume patient prescription and billing data workloads.
- Consolidated data from **Azure Blob**, **ADLS**, **AWS S3**, and on-prem systems while supporting diverse formats such as **JSON**, **XML**, **HL7**, **CSV**, **HTML** and **Parquet**.
- Improved performance and **reduced cloud costs by ~10%** through intelligent **cluster tuning**, **caching strategies**, and **query optimization**.
- Implemented advanced **Spark optimization** practices—including broadcast joins and multithreading—achieving **~20% faster processing** for daily workloads of over 1M records.
- Powered **GenAI** and **ML** initiatives by delivering clean, curated datasets for predictive analytics, including claim-rejection models and IoT-driven equipment maintenance insights.
- Partnered with Agile teams to shape business requirements into scalable data-engineering solutions.
- Continuously monitored industry trends in data engineering and GenAI, advising leadership on adoption strategies.

Project: Legacy System Modernization & Data Migration to Azure

Client: CVS Health | **Duration:** Oct 2020 – Dec 2022 (2 years 3 months)

Role : Data Engineer

Key Contributions:

- Leveraged Azure Data Factory to process data from pharmaceutical packaging machine sensors for predictive maintenance, **lowering unscheduled maintenance incidents by 20%**.
- Re-engineered and migrated critical healthcare data from PICK OS and Mainframe into Azure-based architecture, achieving **job runtime reduction of ~12%** for batch jobs.
- Designed data ingestion and transformation frameworks using Python and Azure Data Factory.
- Ensured compliance with PHI/HIPAA regulations throughout data transformation and migration processes.
- Enabled cross-platform data availability to support reporting, analytics, and downstream API services.

PREVIOUS PROJECTS

Project: OmniDX LTC Workflow – Application Development, Maintenance & Re-engineering

Client: CVS Health | **Duration:** Feb 2017 – Sep 2020 (3 years 8 months)

Role : Technical Architect

Key Contributions

- Architected and implemented solutions for Omnicare's Long-Term Care (LTC) Workflow platform, ensuring alignment with enterprise architecture.
- Migrated and optimized 1.2M+ records of PICK OS into Azure with **cost savings of ~10%** in infrastructure utilization.
- Led end-to-end system design, development, and integration for patient, facility, physician, billing, and inventory modules.
- Designed and executed data migration strategies, enabling **seamless cutover with <1% downtime** and ensuring **99.9% data integrity**.

Project: TAP – Accounts Payable Match and Pharmacy PDX-EPS

Client: Target Corporation | **Duration:** Sep 2013 – Jan 2017 (3 years 5 months)

Role: Senior Software Developer

Key Contributions:

- Developed and optimized AP invoice matching workflows, automating exception handling and vendor reconciliation.
- Built middleware services for secure integration with financial systems and vendors.
- Improved average transaction processing time by **~12%** and reduced manual intervention by **~15%**.
- Implemented electronic prescription workflows, integrating pharmacy systems with Target's ERP.
- Optimized batch jobs for high-volume prescription processing.

Project: Tririga Capital Project Management (TCPM)

Client: Target Corporation | **Duration:** Apr 2012 – Aug 2013 (1 year 5 months)

Role: Senior Software Developer

Key Contributions:

- Customized TCPM modules for capital project planning, budgeting, and approval workflows.
- Reduced reporting cycle time by **~12%** and improved budget tracking accuracy to **~95%**.

Projects: SAP Migration, Financial Data Management, Basic Billing System & Bank Reconciliation Process

Client: Target Corporation | **Duration:** Jul 2006 – Mar 2012 (5 years 9 months)

Role: Software Developer

Key Contributions:

- Enhanced and migrated data for Legacy ERP to SAP FICO.
- Automated batch jobs, implemented exception handling, and optimized EDI vendor interfaces.
- Reduced manual bank reconciliation effort by **~15%** and improved batch runtime efficiency by **~10%**.

EDUCATION

- **B.E - Electronics and Communication** | Birla Institute of Technology, Mesra, Ranchi, India, 2006

CERTIFICATIONS

- Databricks Data Engineer Professional
- Microsoft Fabric Data Engineer Associate (DP-700)
- Google Associate Cloud Engineer (ACE)
- Microsoft Azure Fundamentals (AZ-900), Azure Administrator (AZ-104)
- Scrum.org Certified Professional Scrum Master and Professional Scrum Product Owner